

Cohort modeling in R

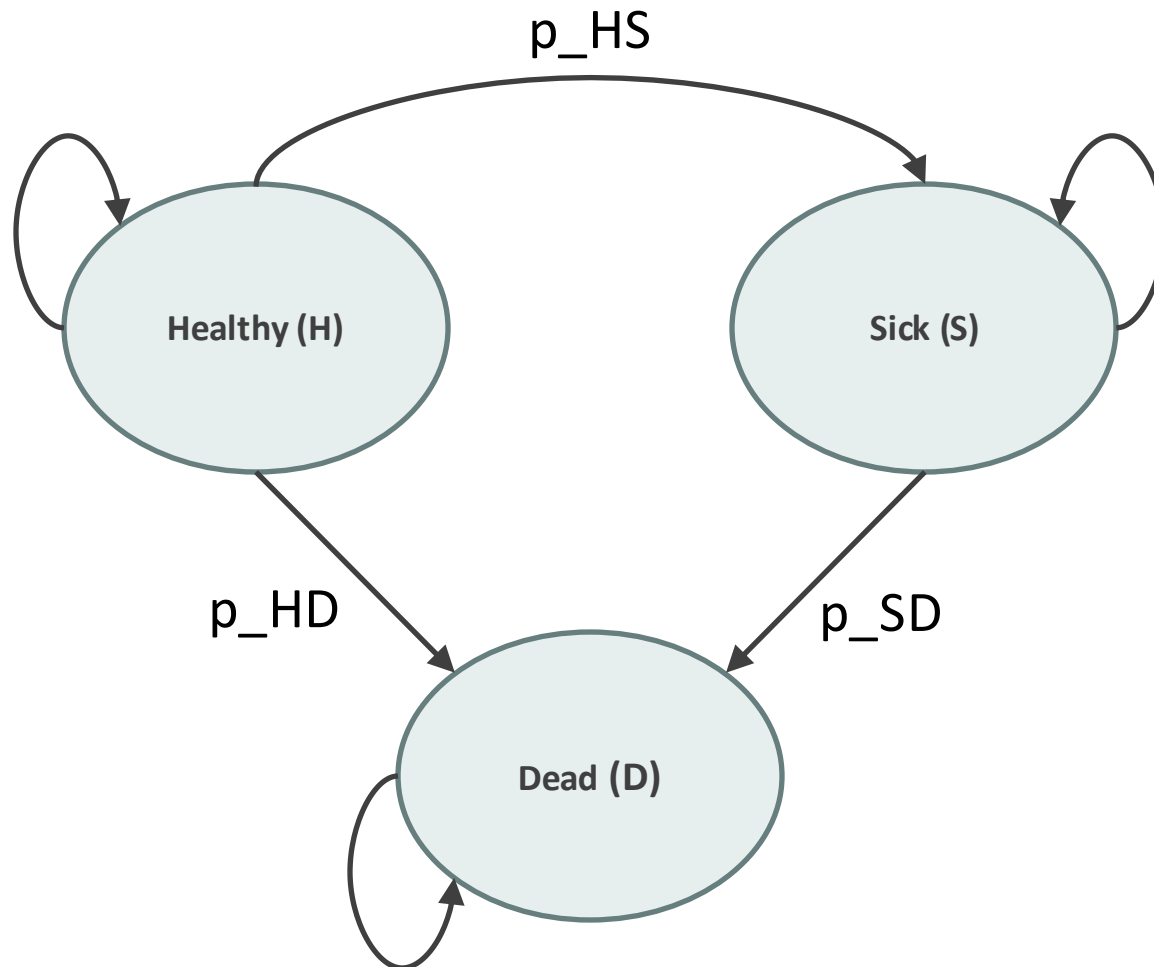
Decision Modeling and Cost-Effectiveness
Analysis in R workshop

August 2025

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Three-State Model



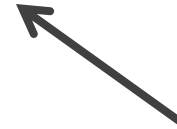
Probabilities vs. Rates

- Probability of an event
 - Likelihood that an event occurs in a fixed amount of time
 - e.g., a person's probability of dying in the next year
- Event rate
 - # of events expected to occur per unit time
 - e.g., number of deaths observed per 1,000 person-years
 - Easily converts to other time units by division
 - e.g., a monthly rate is an annual rate divided by 12
- Can calculate the probability of an event by "accumulating" the event rate over time

Probability <-> Rate Conversion

- Converting a rate to a probability

$$p = 1 - \exp(-r * dt)$$



Length of time interval

- Converting a probability to a rate

$$r = -\ln(1-p) / dt$$

Conversion Examples

- The probability of Event A in a year is 0.95. What is the annual rate?

$$-\ln(1-0.95)/1 = 2.81$$

- The probability of Event B in a year is 0.001. What is the annual rate?

$$-\ln(1-0.001)/1 = 0.001$$

- The rate of Event C occurring is 0.11 per month. What is the probability of that event happening after one year?

$$1-\exp(-0.11*12) = 0.73$$

R Conversion Functions

- In R, we will use the following functions from the `darthtools` R package:

```
rate_to_prob(r = rate, t = cycle_length)
```

```
prob_to_rate(p = probability, t = cycle_length)
```

Why use rates?

- The transition matrix for a discrete time model will always contain transition *probabilities*
 - So why do we need to understand rates?
- A rate may be what's estimated from a study (e.g., from survival analysis)
- Easily adjust rate parameters to appropriate model cycle lengths (just division)
- When modeling disease processes as a *relative increase in risk over a baseline*, using rates avoids invalid probabilities

Relative parameterization example

- For 80 year-olds, annual baseline rate of dying if healthy is 0.044 in the US (0.037 in Canada)
- With Disease X, rate of death is 100x higher
- An 80 year-old with Disease X has an annual rate of dying of 4.4 (3.7 in Canada)
 - Annual probability of dying with Disease X is $1 - \exp(-4.4) = 0.98$